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COP 6819 — Advanced Internet Systems

Cloud Analyst Assignment

Summary of Results:

I ran six CloudAnalyst simulations. The first five use the same base topology (6 user bases, 4 data centers, 1,000 peak users per base at 60 requests/hour) and change one thing at a time to see how load-balancing policy and data-center provisioning affect performance and cost: a Round Robin baseline, a Throttled run, an under-provisioned data center (VM-scheduling edge case), a network-latency edge case, and a cost-sensitive run with large payloads. The sixth run tests the Equally Spread Current Execution Load policy on a larger map of 7 user bases and 6 data centers with custom network latencies.

The headline finding is that overall response time barely moves — the five runs on the shared topology all land between 92 and 95 ms — because inter-region network latency dominates the total and data-center processing time is only a few milliseconds at most. The interesting differences show up in two places the average hides: the worst-case (maximum) response time, which is driven by under-provisioning, and the total cost, which is driven far more by data-transfer volume than by the number of VMs. UB5 sits ~300 ms away from its data center in runs 1-5; that is a fixed geographic-latency floor, not a load-balancing effect, and it appears identically across those scenarios. The sixth run shows the same pattern on a different map: processing stays under 1 ms on average, response time tracks distance, and a data center that receives no traffic still adds to the bill.

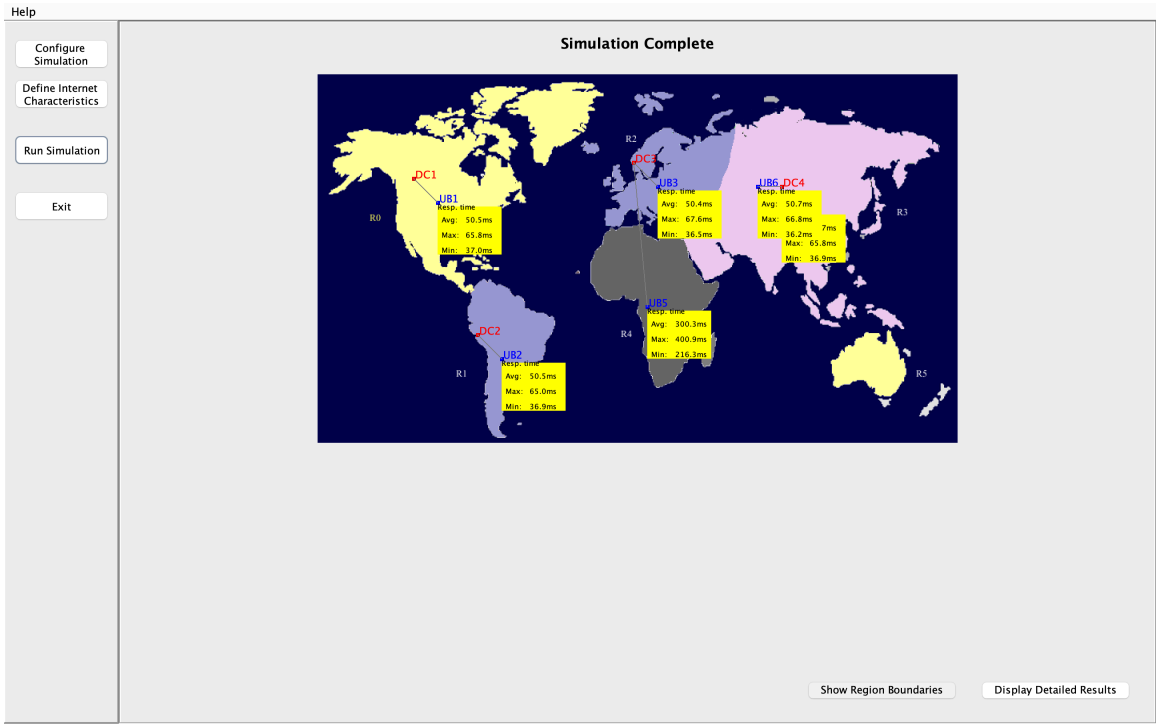
Explanation of the scenarios:

- **Round Robin (baseline):** the service broker hands each new request to the next VM in order (0, 1, 2, … N, 0, 1, …). All four data centers are well provisioned.
- **Throttled:** the broker sends each request to the next available VM and caps the number of simultaneous requests per VM; here three data centers run on a single VM each.
- **Under-provisioned DC (VM-scheduling edge case):** the same starved layout, examined for worst-case tail latency instead of the average.
- **Network / Internet latency edge case:** only one data center (DC2) is left weak to isolate a single regional bottleneck.

- **Cost-sensitive:** compute is generous but two user bases send 100× larger requests to expose data-transfer cost.
- **Equally Spread Current Execution Load (load-based):** the load balancer sends each request to the VM with the fewest active requests. Run on a larger map of 7 user bases and 6 data centers with custom network latencies.

Round Robin Results:

Baseline scenario. Four data centers provisioned with 15 / 5 / 5 / 30 VMs (DC1-DC4) and the Round Robin service-broker policy. This is the reference every other scenario is compared against.



Results of the Simulation Completed at: 21/09/2026 00:48:19

Overall Response Time Summary

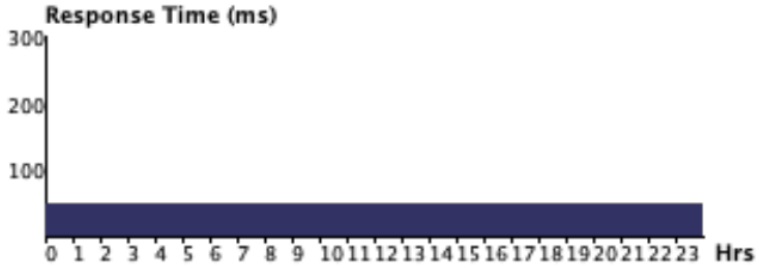
	Avg (ms)	Min (ms)	Max (ms)
Overall response time:	92.59	36.19	400.88
Data Center processing time:	0.84	0.03	3.40

Response Time by Region

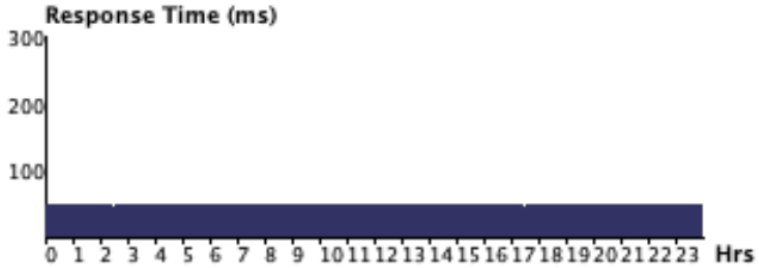
Userbase	Avg (ms)	Min (ms)	Max (ms)
UB1	50.54	37.02	65.75
UB2	50.52	36.92	65.01
UB3	50.43	36.54	67.56
UB4	50.70	36.89	65.76
UB5	300.27	216.26	400.88
UB6	50.70	36.19	66.77

User Base Hourly Response Times

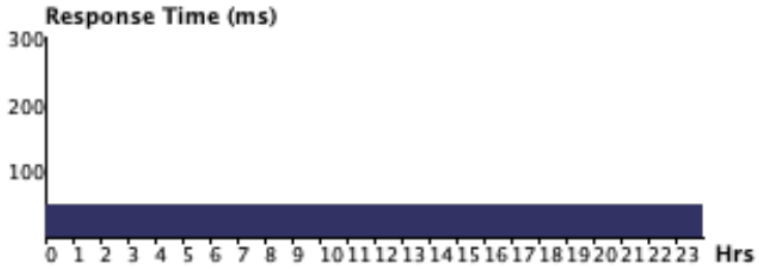
UB1



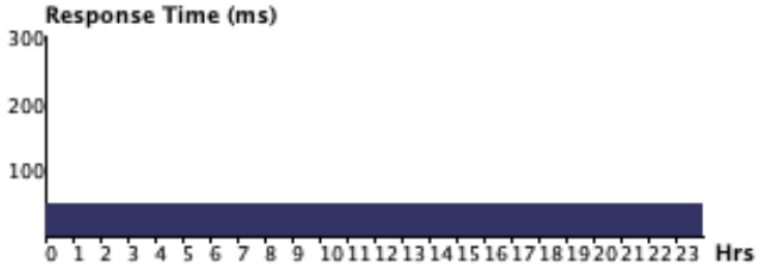
UB2



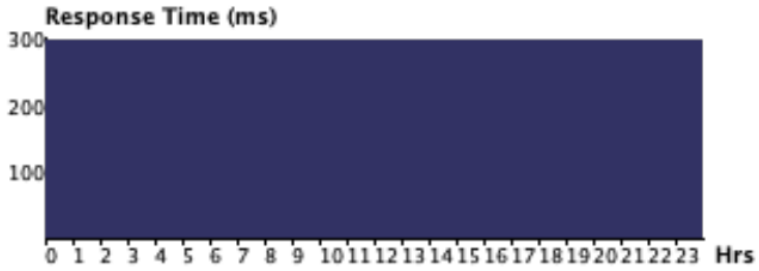
UB3



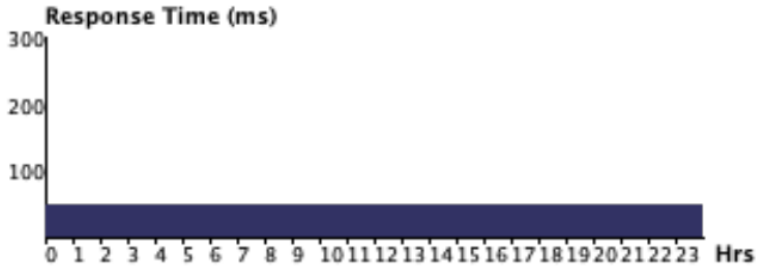
UB4



UB5



UB6

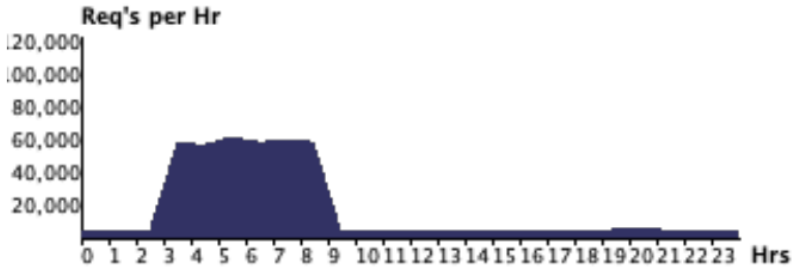


Data Center Request Servicing Times

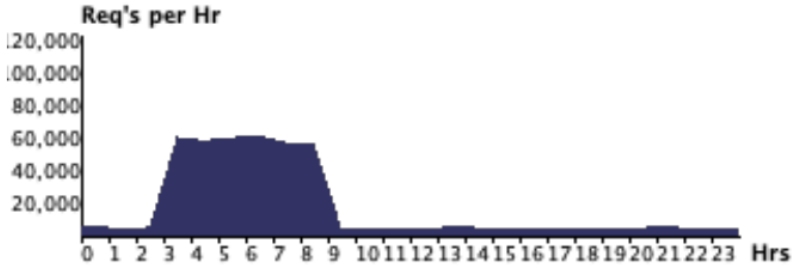
Data Center	Avg (ms)	Min (ms)	Max (ms)
DC1	0.88	0.05	2.90
DC2	0.80	0.03	3.40
DC3	0.67	0.03	3.25
DC4	1.02	0.07	2.13

Data Center Hourly Loading

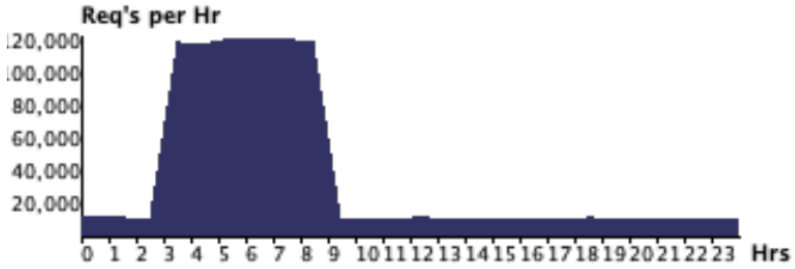
DC1



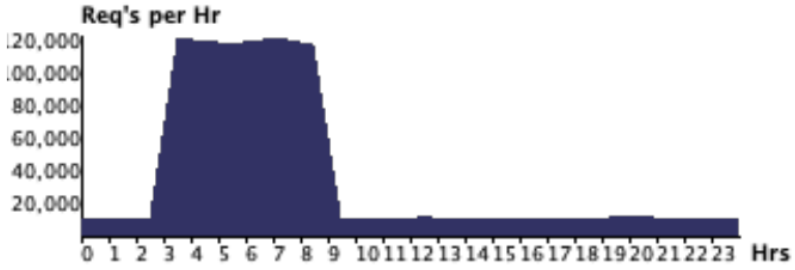
DC2



DC3



DC4



Cost

Total Virtual Machine Cost ($):	**132.01**
Total Data Transfer Cost ($):	29.35
Grand Total ($):	161.36

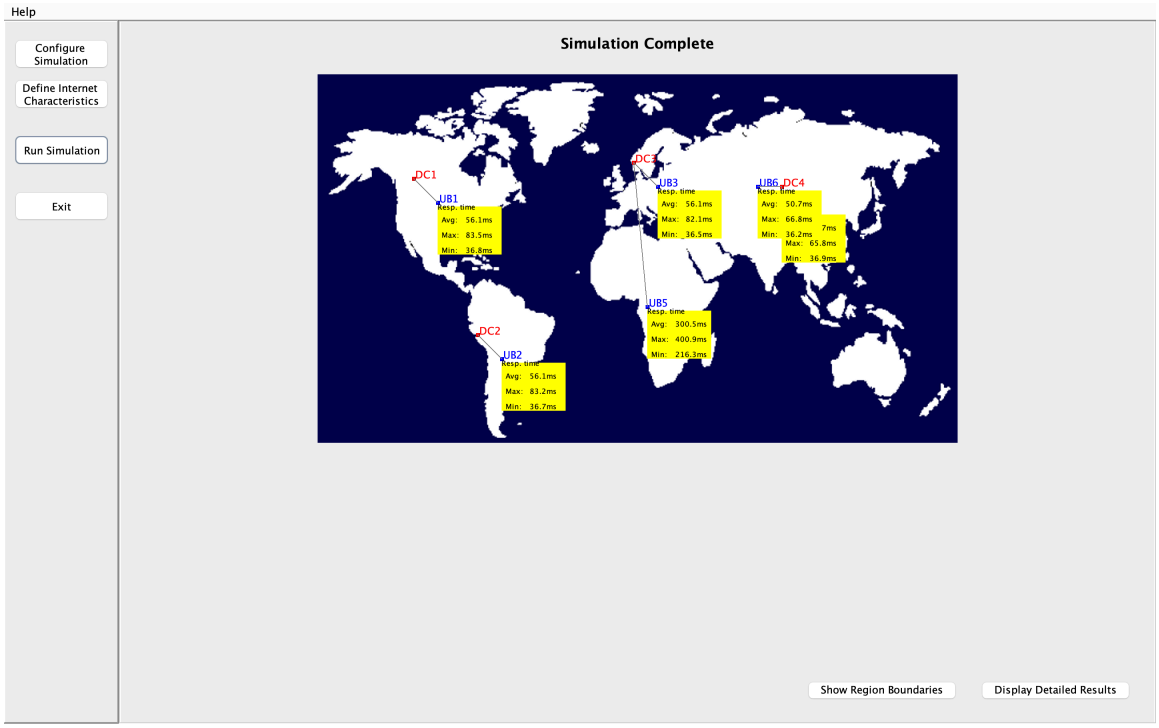
Data Center	VM Cost $	Data Transfer Cost $	Total $
DC2	12.00	4.88	16.88
DC1	36.00	4.87	40.87
DC4	72.01	9.78	81.78
DC3	12.00	9.82	21.82

Analysis

With every data center adequately provisioned, Round Robin spreads requests evenly and gives the lowest overall response time of the five runs (92.59 ms) with essentially free processing time (0.84 ms). No data center is stressed — all hourly processing charts stay flat. DC4 dominates the bill ($72) because it holds 30 VMs.

Throttled Results:

Throttled service-broker policy with DC1-DC3 deliberately starved to 1 VM each while DC4 keeps 30 (1 / 1 / 1 / 30). Tests what the throttle does when compute is scarce.



Results of the Simulation Completed at: 21/09/2026 01:07:38

Overall Response Time Summary

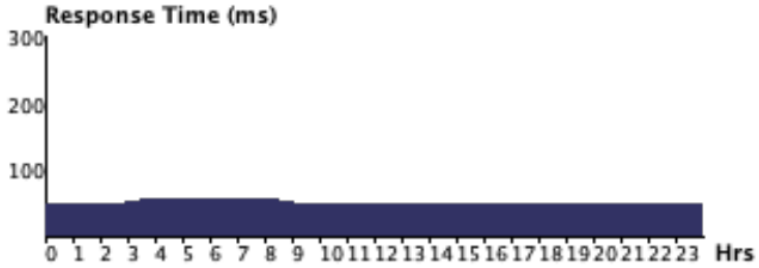
	Avg (ms)	Min (ms)	Max (ms)
Overall response time:	95.43	36.19	400.88
Data Center processing time:	3.68	0.03	23.76

Response Time by Region

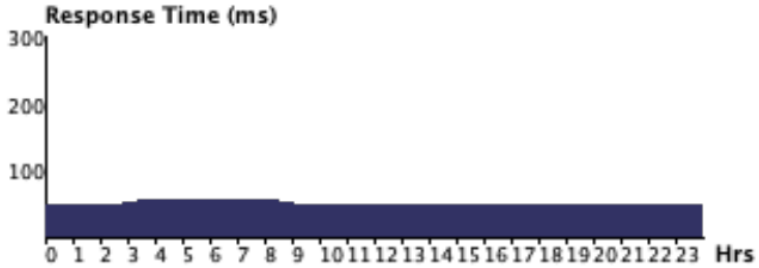
Userbase	Avg (ms)	Min (ms)	Max (ms)
UB1	56.08	36.84	83.54
UB2	56.10	36.72	83.21
UB3	56.08	36.54	82.07
UB4	50.70	36.89	65.76
UB5	300.53	216.26	400.88
UB6	50.70	36.19	66.77

User Base Hourly Response Times

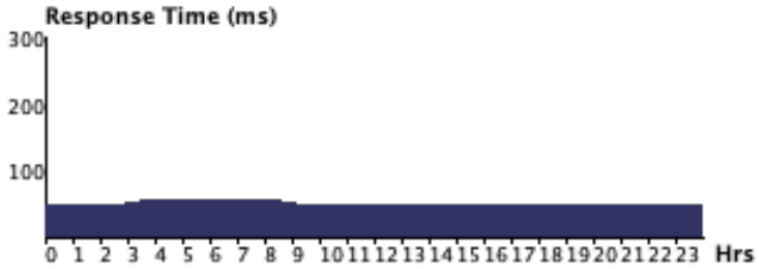
UB1



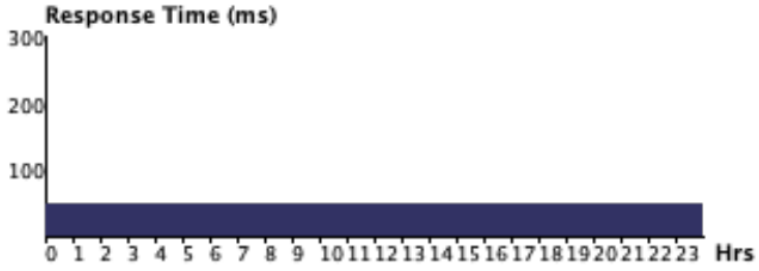
UB2



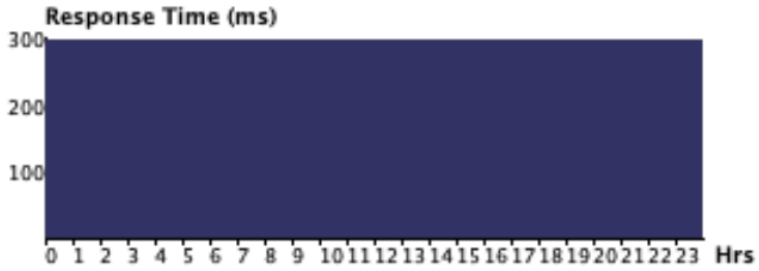
UB3



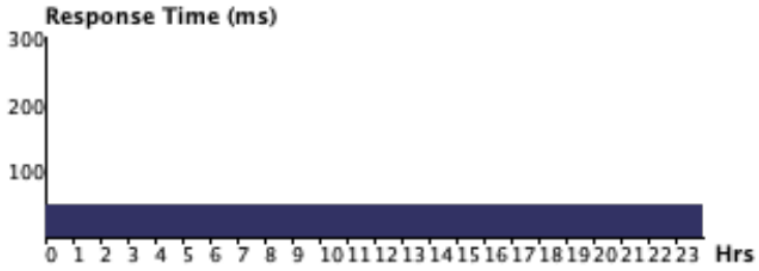
UB4



UB5



UB6

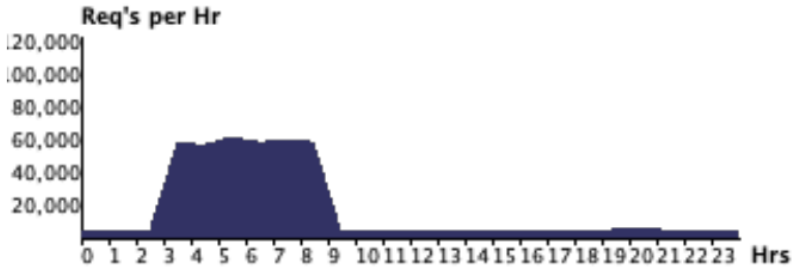


Data Center Request Servicing Times

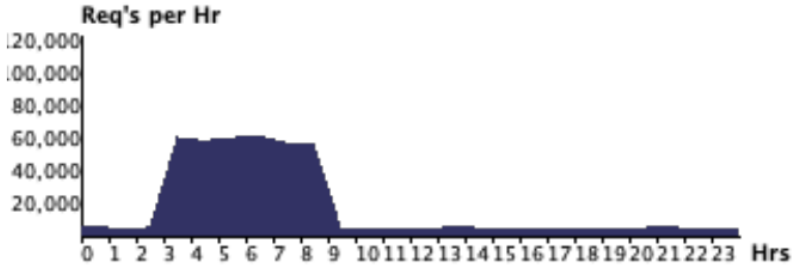
Data Center	Avg (ms)	Min (ms)	Max (ms)
DC1	6.43	0.03	23.76
DC2	6.41	0.03	22.10
DC3	3.61	0.03	23.29
DC4	1.02	0.07	1.51

Data Center Hourly Loading

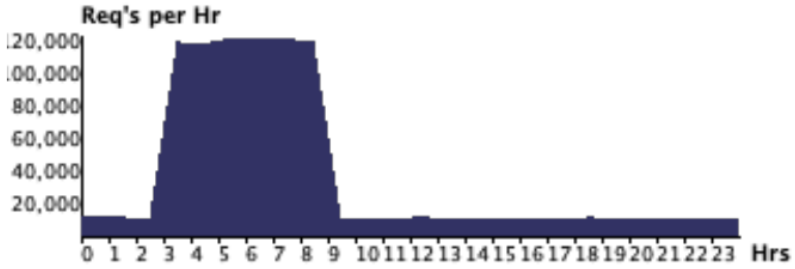
DC1



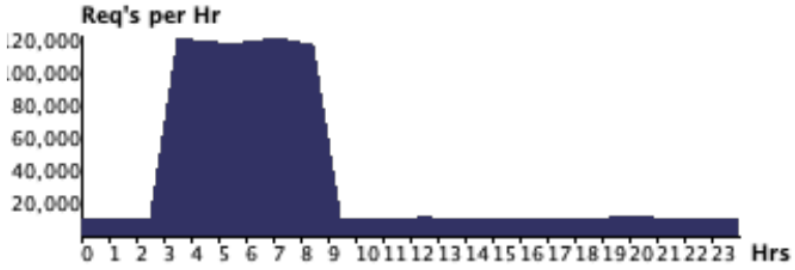
DC2



DC3



DC4



Cost

Total Virtual Machine Cost ($):	**79.21**
Total Data Transfer Cost ($):	29.35
Grand Total ($):	108.56

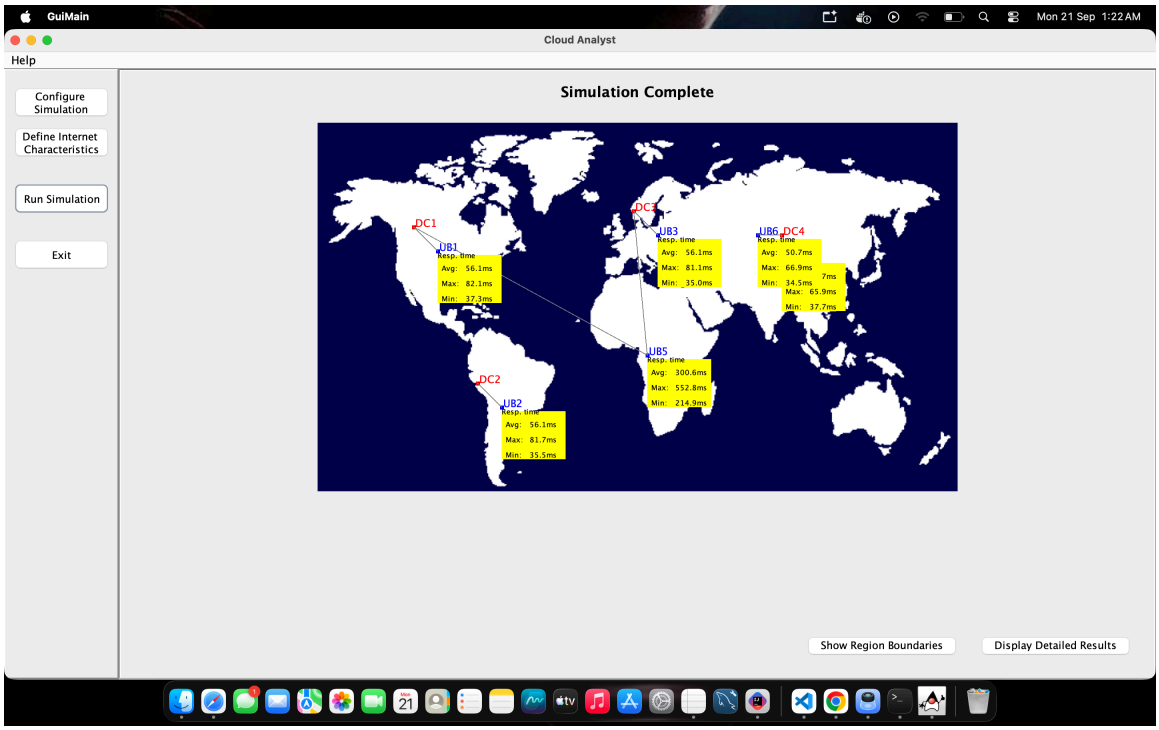
Data Center	VM Cost $	Data Transfer Cost $	Total $
DC2	2.40	4.88	7.28
DC1	2.40	4.87	7.27
DC4	72.01	9.78	81.78
DC3	2.40	9.82	12.22

Analysis

Throttling requests onto a single VM per data center pushes average processing time from 0.84 ms up to 3.68 ms (DC1/DC2 climb to ~6.4 ms) as requests queue. Overall response only rises to 95.43 ms because processing is still tiny next to network latency. The payoff is cost: $108.56, the cheapest run, because far fewer VMs are rented.

Under-Provisioned Data Center Results (VM-Scheduling Edge Case):

Same 1 / 1 / 1 / 30 layout as the Throttled run, re-examined to expose worst-case (tail) latency rather than the average when a single-VM data center cannot schedule a burst fast enough.



Results of the Simulation Completed at: 21/09/2026 01:21:56

Overall Response Time Summary

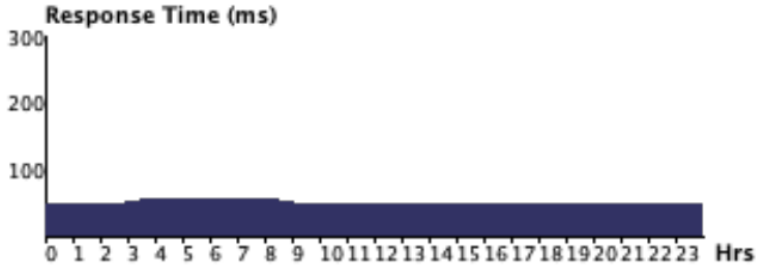
	Avg (ms)	Min (ms)	Max (ms)
Overall response time:	95.44	34.47	552.75
Data Center processing time:	3.68	0.03	24.26

Response Time by Region

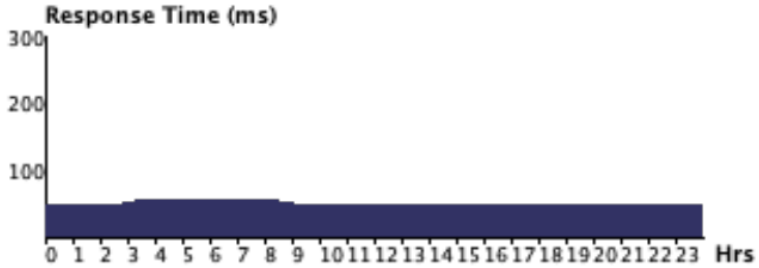
Userbase	Avg (ms)	Min (ms)	Max (ms)
UB1	56.06	37.34	82.06
UB2	56.13	35.52	81.66
UB3	56.09	35.01	81.08
UB4	50.66	37.71	65.88
UB5	300.64	214.88	552.75
UB6	50.70	34.47	66.89

User Base Hourly Response Times

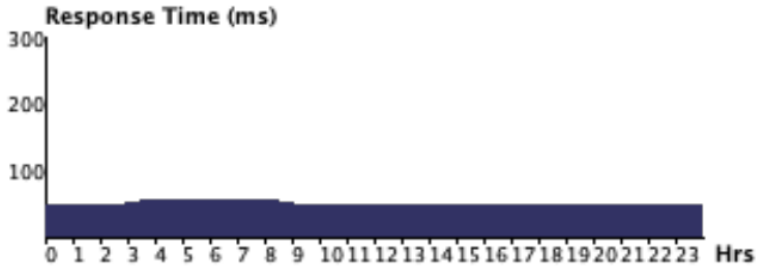
UB1



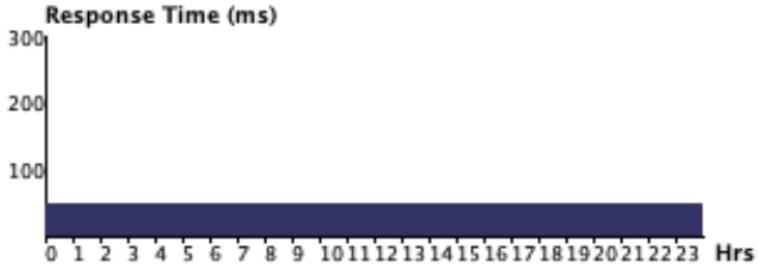
UB2



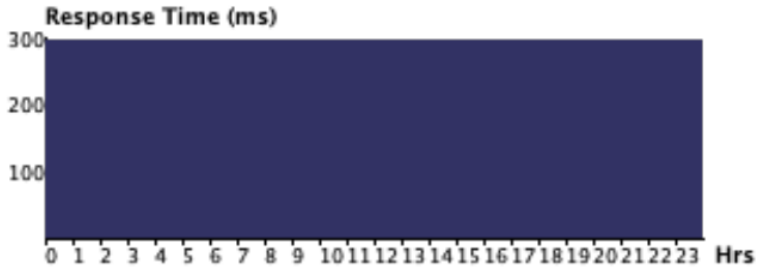
UB3



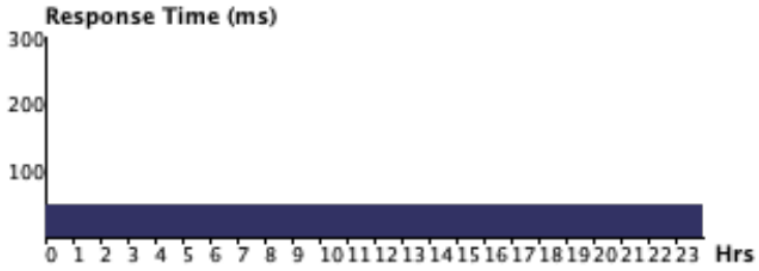
UB4



UB5



UB6

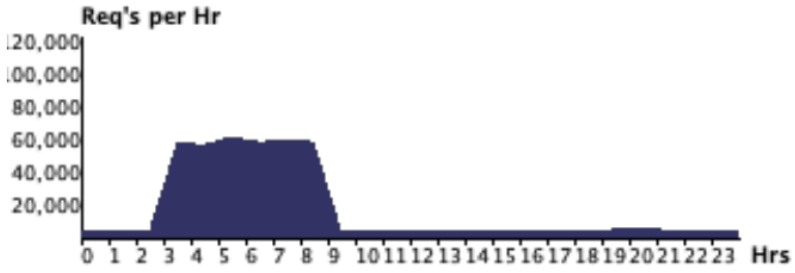


Data Center Request Servicing Times

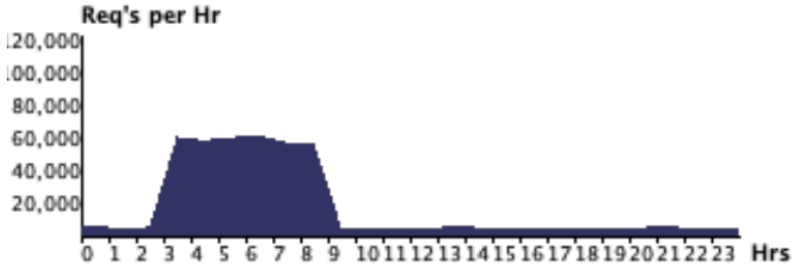
Data Center	Avg (ms)	Min (ms)	Max (ms)
DC1	6.42	0.03	24.26
DC2	6.43	0.03	21.79
DC3	3.61	0.03	23.26
DC4	1.02	0.07	1.50

Data Center Hourly Loading

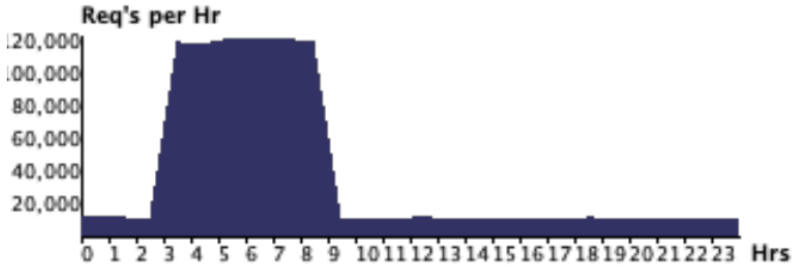
DC1



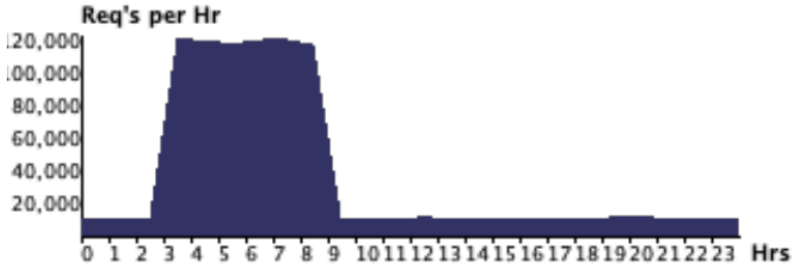
DC2



DC3



DC4



Cost

Total Virtual Machine Cost ($):	**79.21**
Total Data Transfer Cost ($):	29.35
Grand Total ($):	108.56

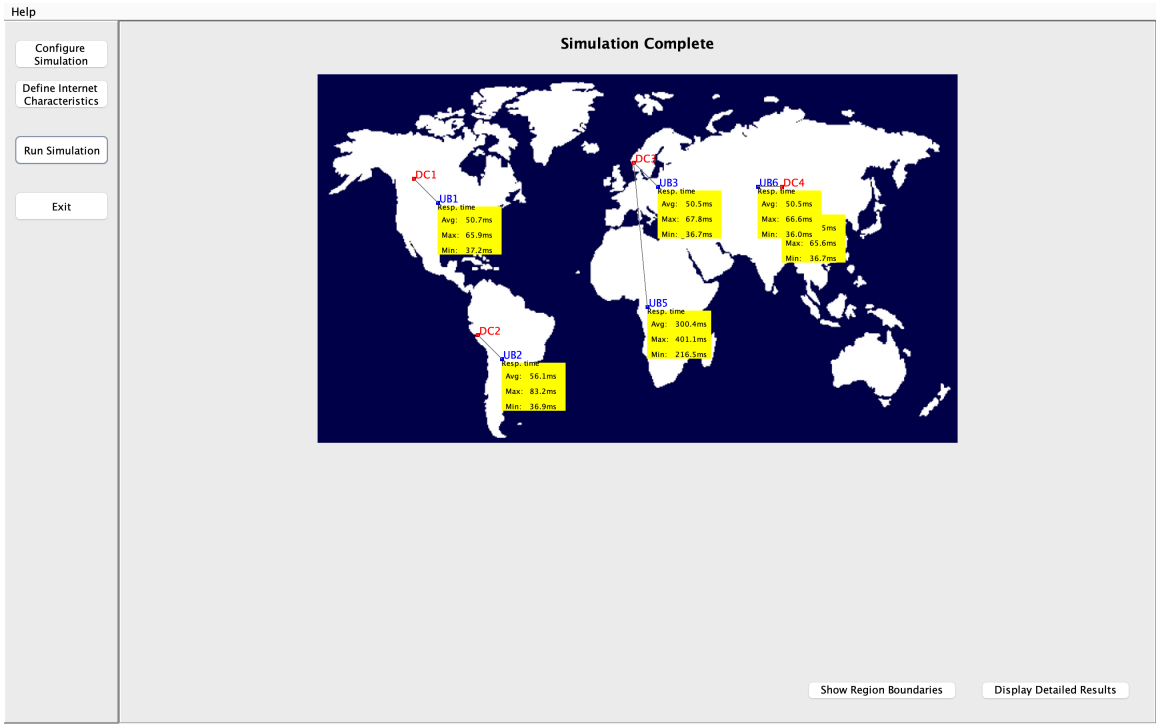
Data Center	VM Cost $	Data Transfer Cost $	Total $
DC2	2.40	4.88	7.28
DC1	2.40	4.87	7.27
DC4	72.01	9.78	81.78
DC3	2.40	9.82	12.22

Analysis

The averages here are almost identical to the Throttled run (95.44 ms, $108.56), but the maximum response time spikes to 552.75 ms — about 38% worse than the ~400 ms seen elsewhere — and DC processing max hits 24.26 ms. This is the key lesson: averages hide under-provisioning; the damage only shows up in the tail, which is exactly what SLA-sensitive workloads care about.

Network / Internet Latency Edge Case Results:

VMs distributed 20 / 1 / 20 / 20 so that only DC2 is under-provisioned. Isolates a single weak data center as a regional bottleneck.



Results of the Simulation Completed at: 21/09/2026 01:32:08

Overall Response Time Summary

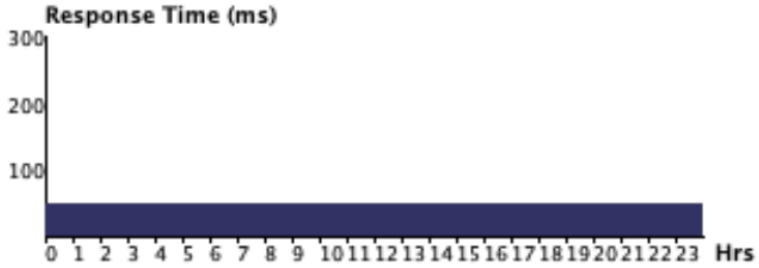
	Avg (ms)	Min (ms)	Max (ms)
Overall response time:	93.51	36.00	401.07
Data Center processing time:	1.76	0.03	22.10

Response Time by Region

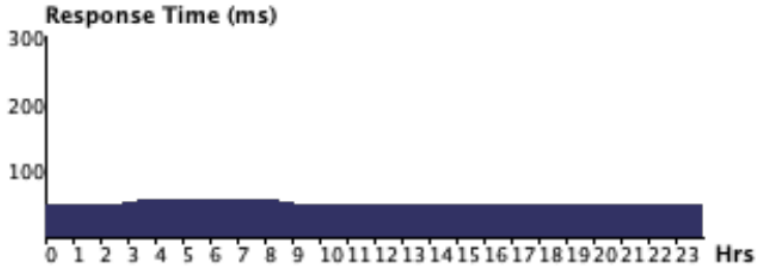
Userbase	Avg (ms)	Min (ms)	Max (ms)
UB1	50.67	37.21	65.94
UB2	56.11	36.92	83.21
UB3	50.46	36.72	67.75
UB4	50.51	36.70	65.57
UB5	300.43	216.45	401.07
UB6	50.50	36.00	66.58

User Base Hourly Response Times

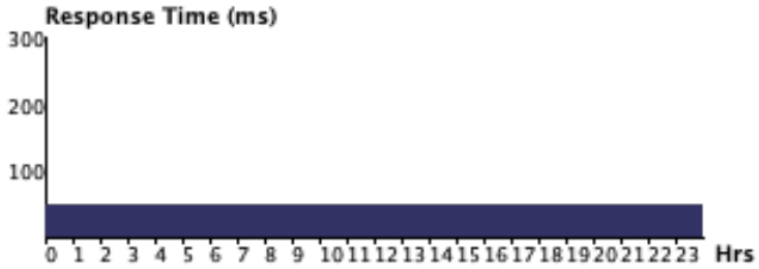
UB1



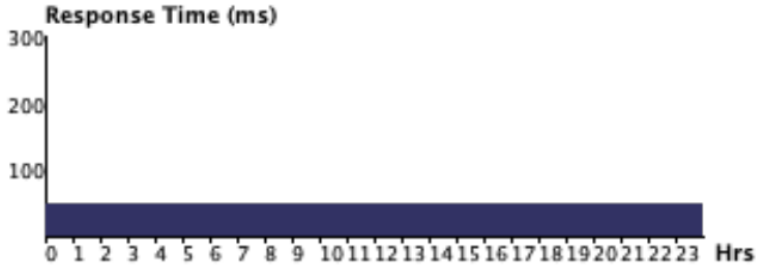
UB2



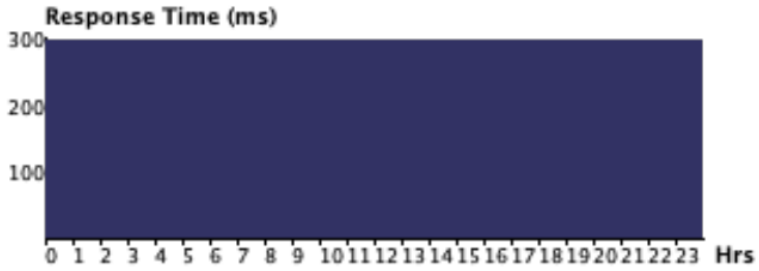
UB3



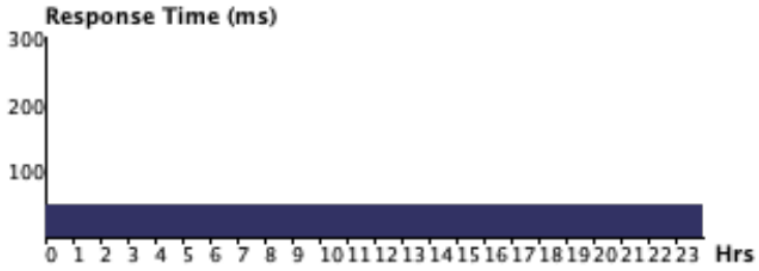
UB4



UB5



UB6

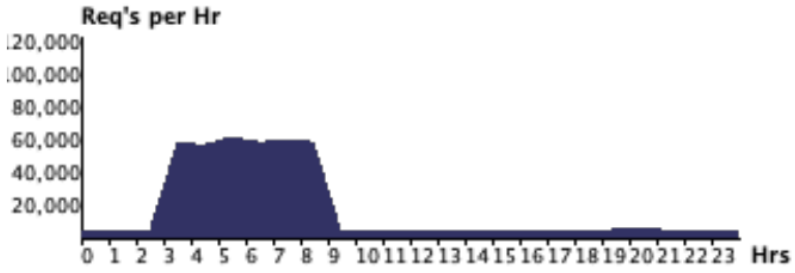


Data Center Request Servicing Times

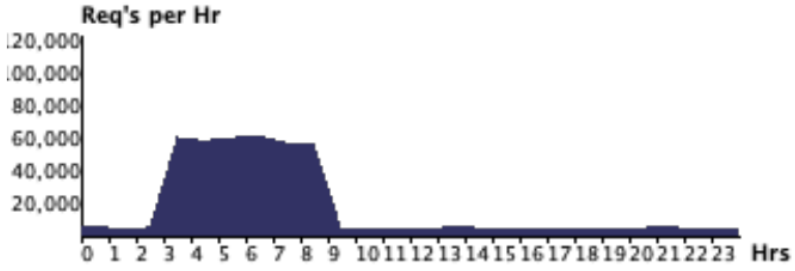
Data Center	Avg (ms)	Min (ms)	Max (ms)
DC1	1.01	0.07	1.53
DC2	6.41	0.03	22.10
DC3	0.76	0.04	1.24
DC4	0.82	0.04	1.32

Data Center Hourly Loading

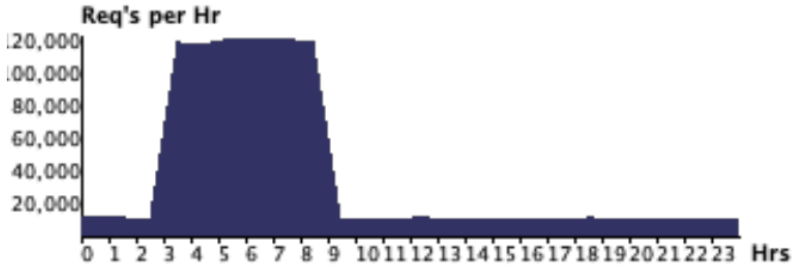
DC1



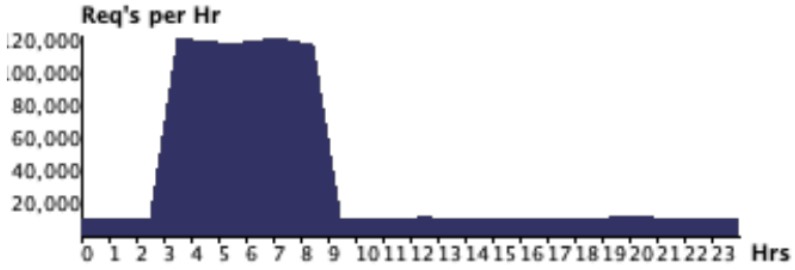
DC2



DC3



DC4



Cost

Total Virtual Machine Cost ($):	**146.42**
Total Data Transfer Cost ($):	29.35
Grand Total ($):	175.76

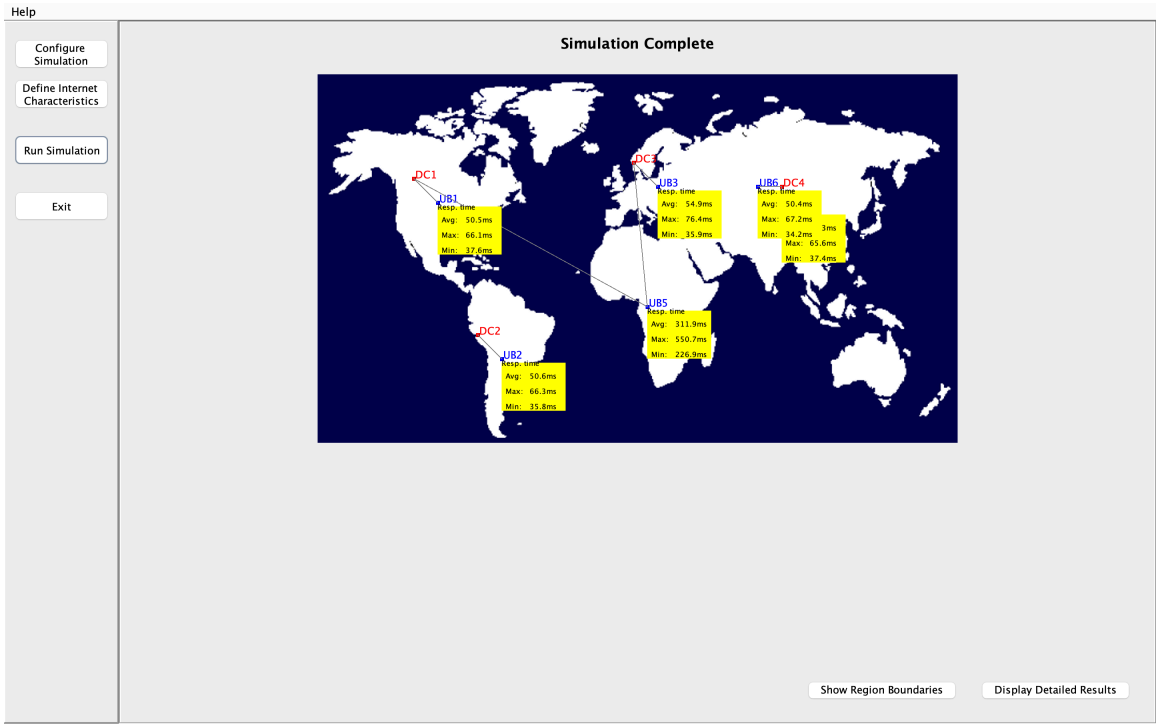
Data Center	VM Cost $	Data Transfer Cost $	Total $
DC2	2.40	4.88	7.28
DC1	48.01	4.87	52.87
DC4	48.01	9.78	57.78
DC3	48.01	9.82	57.83

Analysis

Only DC2 is weak, and the result isolates it cleanly: DC2's average processing time is 6.41 ms while the other three sit near 1 ms, and UB2 (served by DC2) rises to 56 ms versus ~50 ms for the rest. Overall response stays near baseline (93.51 ms) because only one region is hurt. Load balancing cannot compensate for a region that simply lacks capacity — the fix is provisioning, not policy.

Cost-Sensitive Results (Large Payloads):

Generously balanced VMs (15 / 15 / 20 / 15), but UB3 and UB5 send 10,000-byte requests (100× the 100-byte baseline). Tests how payload size drives cost rather than compute.



Results of the Simulation Completed at: 21/09/2026 01:47:10

Overall Response Time Summary

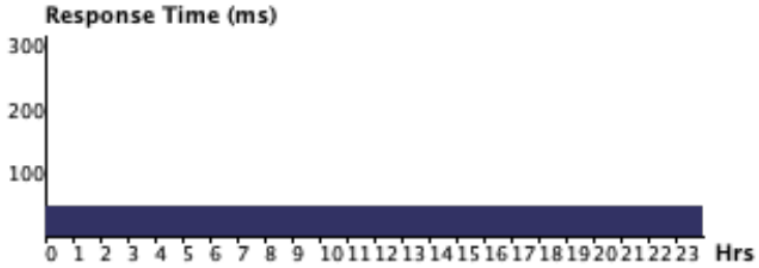
	Avg (ms)	Min (ms)	Max (ms)
Overall response time:	95.20	34.16	550.67
Data Center processing time:	0.84	0.04	1.55

Response Time by Region

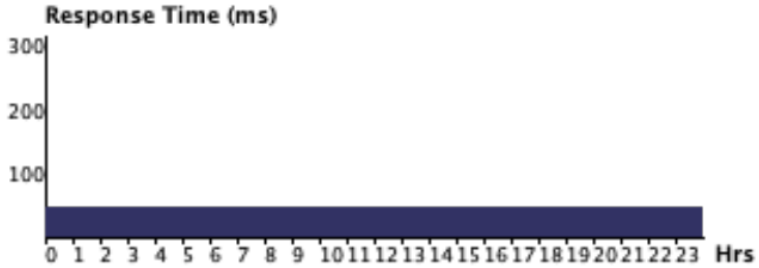
Userbase	Avg (ms)	Min (ms)	Max (ms)
UB1	50.51	37.59	66.06
UB2	50.62	35.77	66.26
UB3	54.90	35.86	76.39
UB4	50.35	37.40	65.57
UB5	311.93	226.88	550.67
UB6	50.39	34.16	67.16

User Base Hourly Response Times

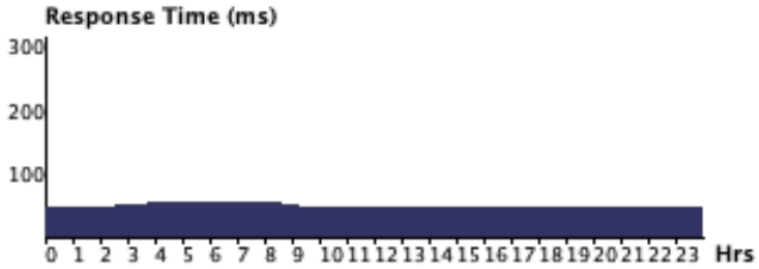
UB1



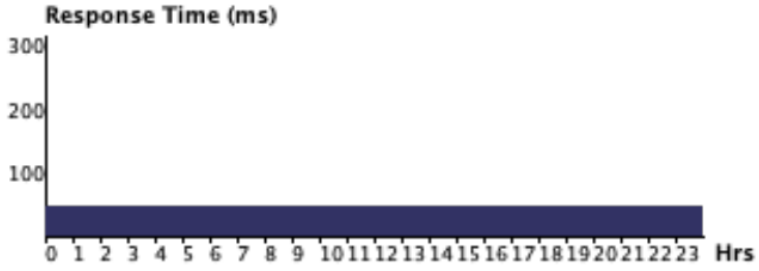
UB2



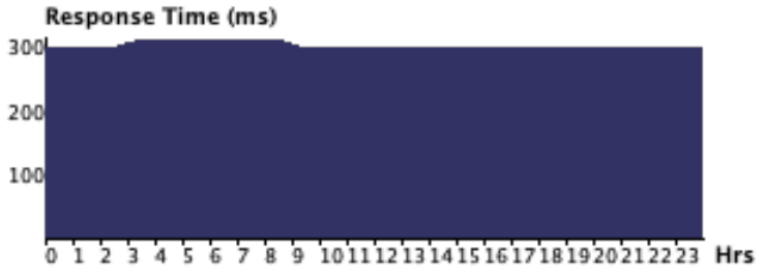
UB3



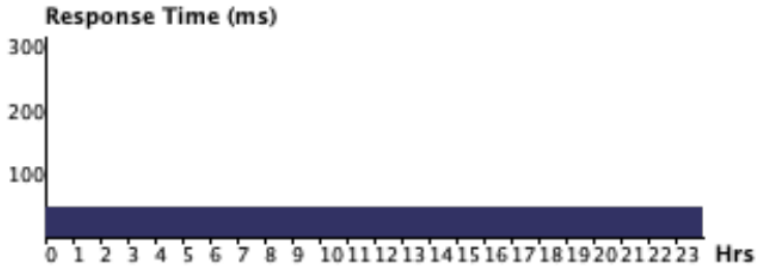
UB4



UB5



UB6

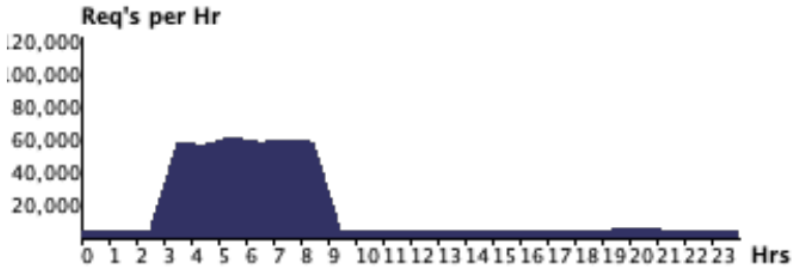


Data Center Request Servicing Times

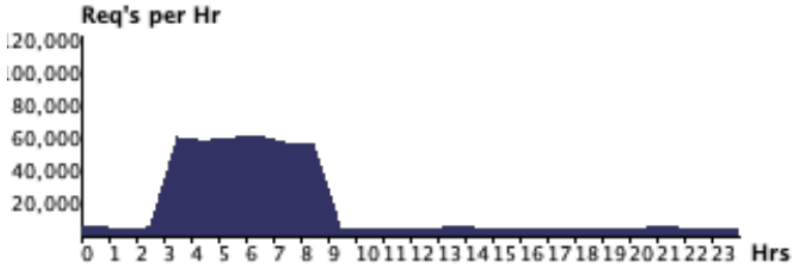
Data Center	Avg (ms)	Min (ms)	Max (ms)
DC1	0.86	0.05	1.40
DC2	0.89	0.06	1.55
DC3	0.92	0.07	1.44
DC4	0.71	0.04	1.24

Data Center Hourly Loading

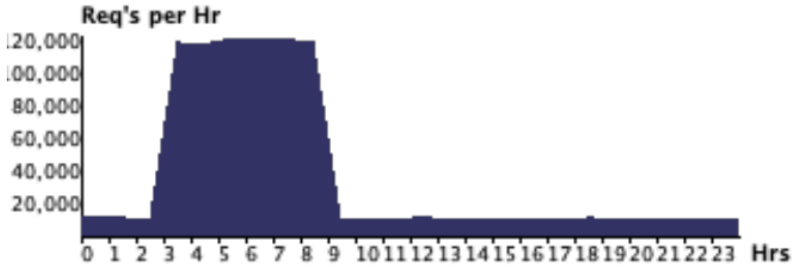
DC1



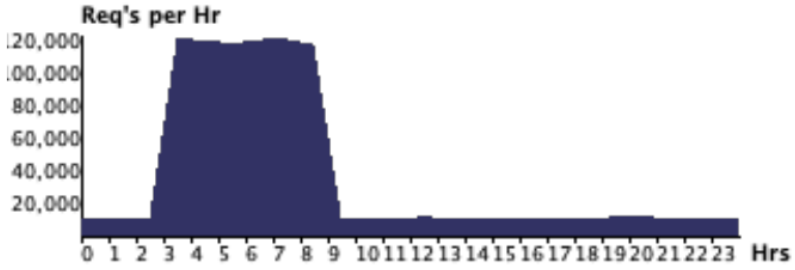
DC2



DC3



DC4



Cost

Total Virtual Machine Cost ($):	**156.02**
Total Data Transfer Cost ($):	1001.58
Grand Total ($):	1157.59

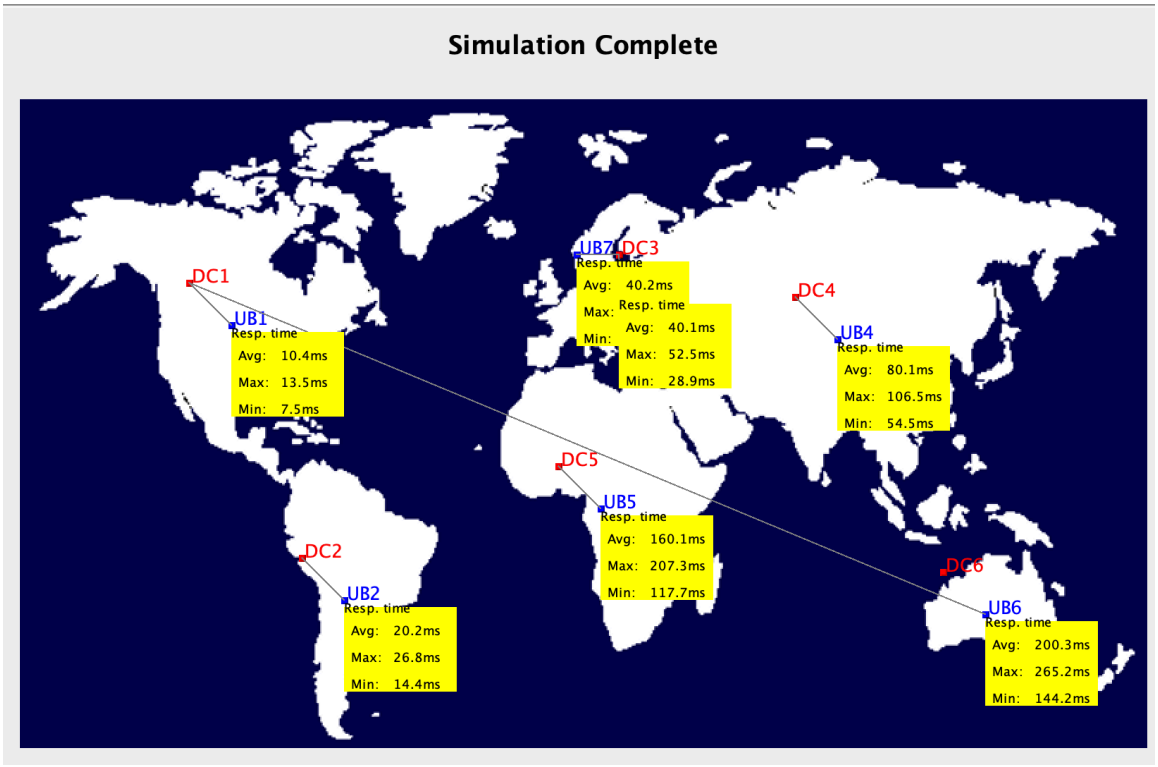
Data Center	VM Cost $	Data Transfer Cost $	Total $
DC2	36.00	4.88	40.89
DC1	36.00	4.93	40.93
DC4	36.00	9.78	45.78
DC3	48.01	981.99	1029.99

Analysis

Plenty of VMs give the best processing time of all runs (0.84 ms avg, 1.55 ms max) and near-empty processing charts — but the large payloads make data-transfer cost explode from ~$29 to $1,001.58, almost all of it at DC3 ($981.99 in transfer alone). Grand total reaches $1,157.59, roughly 7× the baseline. In cloud pricing, data transfer, not compute, is often the dominant cost.

Equally Spread Current Execution Load Results (Load-Based):

Equally Spread Current Execution Load policy on a larger map: 7 user bases and 6 data centers with 5 / 5 / 5 / 5 / 5 / 4 VMs (DC1-DC6) and custom network latencies. Tests whether the load balancer keeps processing flat when some data centers carry twice the traffic of others.



Screenshot from a rerun of the same setup at 21:46 the same day, with the user bases entered in a different order: the rerun's UB4, UB5, UB6 and UB7 are UB5, UB6, UB7 and UB4 in the tables below. Average response times agree within 0.1 ms; minimums and maximums vary by a few milliseconds between runs. The tables come from the original run.

Results of the Simulation Completed at: 28/09/2026 18:14:46

Overall Response Time Summary

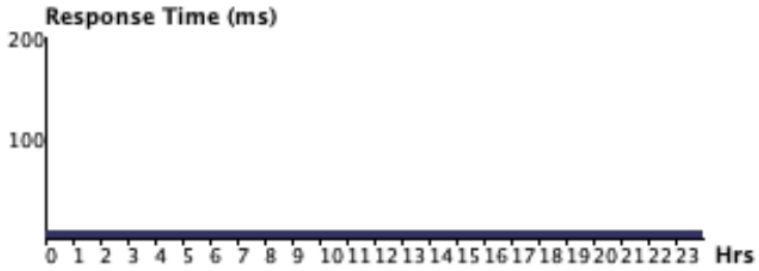
	Avg (ms)	Min (ms)	Max (ms)
Overall response time:	78.81	7.47	258.31
Data Center processing time:	0.55	0.00	2.46

Response Time by Region

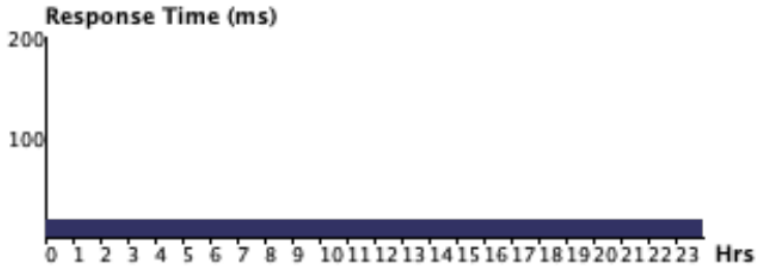
Userbase	Avg (ms)	Min (ms)	Max (ms)
UB1	10.42	7.47	13.46
UB2	20.23	14.43	26.84
UB3	40.12	28.94	52.53
UB4	40.15	27.34	53.35
UB5	80.14	58.93	103.74
UB6	160.17	115.34	212.14
UB7	200.23	149.12	258.31

User Base Hourly Response Times

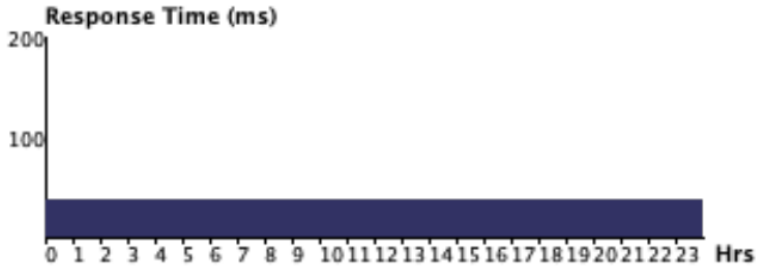
UB1



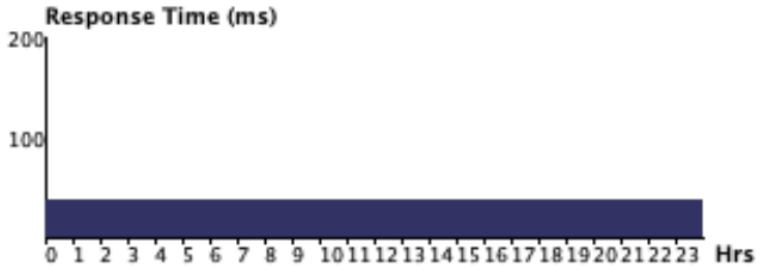
UB2



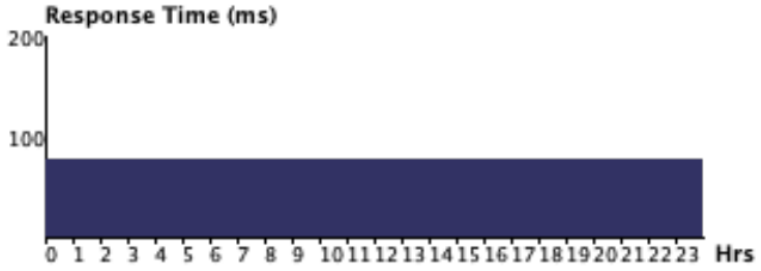
UB3



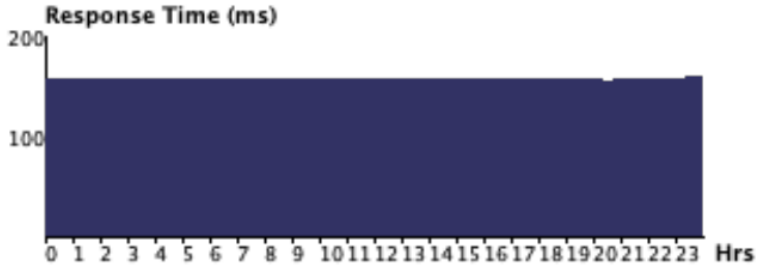
UB4



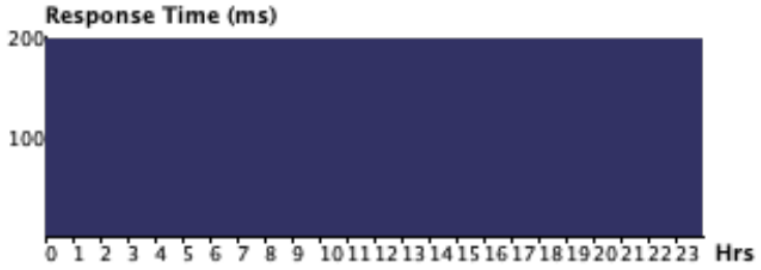
UB5



UB6



UB7

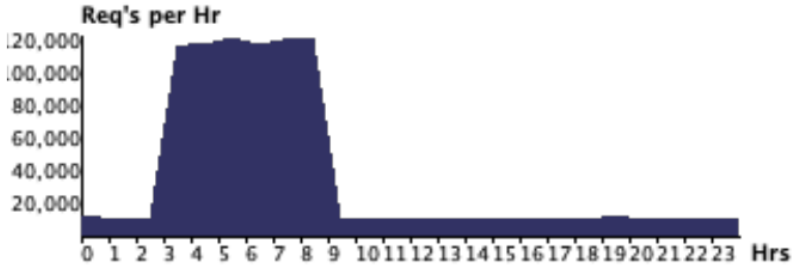


Data Center Request Servicing Times

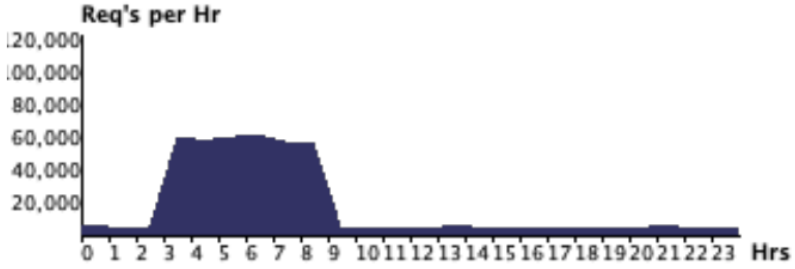
Data Center	Avg (ms)	Min (ms)	Max (ms)
DC1	0.51	0.01	2.46
DC2	0.67	0.01	1.76
DC3	0.54	0.01	1.51
DC4	0.53	0.01	1.18
DC5	0.54	0.01	1.06
DC6	0.00	0.00	0.00

Data Center Hourly Loading

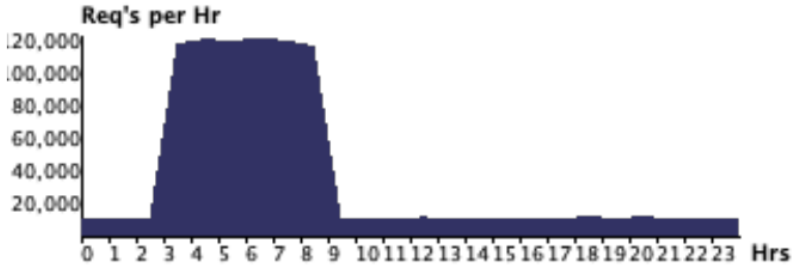
DC1



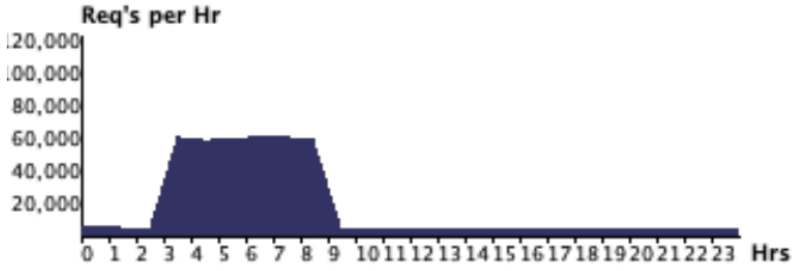
DC2



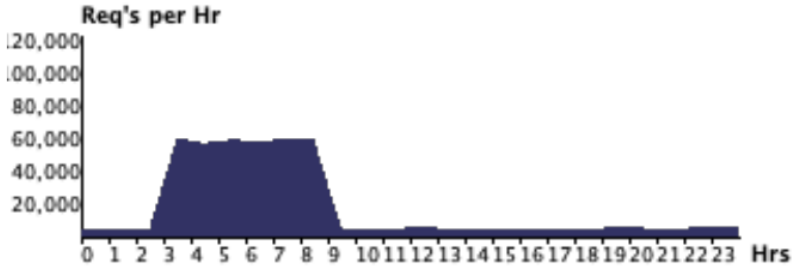
DC3



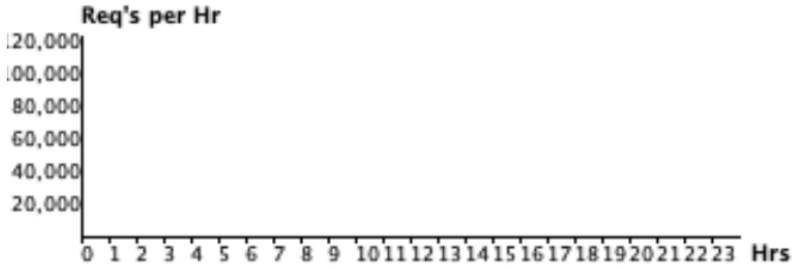
DC4


Req's per Hr
120,000
100,000
80,000
60,000
40,000
20,000
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
Hrs

DC5


Req's per Hr
120,000
100,000
80,000
60,000
40,000
20,000
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
Hrs

DC6


Req's per Hr
120,000
100,000
80,000
60,000
40,000
20,000
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
Hrs

Cost

Total Virtual Machine Cost ($):	**69.61**
Total Data Transfer Cost ($):	34.24
Grand Total ($):	103.85

Data Center	VM Cost $	Data Transfer Cost $	Total $
DC2	12.00	4.88	16.88
DC1	12.00	9.76	21.76
DC4	12.00	4.94	16.94
DC3	12.00	9.78	21.79
DC6	9.60	0.00	9.60
DC5	12.00	4.87	16.88

Analysis

Equally Spread sends each request to the VM with the fewest active requests, and processing is the flattest of all six runs: 0.55 ms average, 2.46 ms maximum. DC1 and DC3 each carry about twice the peak traffic of DC2, DC4 and DC5 (about 120,000 vs 60,000 requests per hour, and double the data-transfer cost), yet their average processing times (0.51 and 0.54 ms) match the others. Response time per user base simply tracks network distance, from 10.42 ms (UB1) to 200.23 ms (UB7). DC6 received no requests but still billed $9.60 for its 4 VMs: the load-balancing policy only picks a VM inside a data center, while the service broker decides which data center gets the traffic. The $103.85 grand total is the lowest of the six runs, but the larger map means it is not a like-for-like comparison with runs 1-5.

Overall Conclusions

- Network latency dominates the average. The five runs on the shared topology cluster at 92–95 ms regardless of policy, because inter-region delay dwarfs data-center processing (≤7 ms). Load-balancing policy mainly moves the small processing component.
- Provisioning drives the tail, not the mean. Starving data centers barely moves the average but sharply raises the maximum response time (up to 552 ms) — tail latency is the real cost of under-provisioning.
- On the shared topology, Throttled is the cheapest ($108.56) and Round Robin the fastest (92.59 ms). The right choice depends on whether the SLA or the budget is the binding constraint.
- Data-transfer cost can dwarf everything. The cost-sensitive run hit $1,157.59, with $1,001.58 of that being data transfer — payload size, not VM count, is the biggest cost lever.
- Load balancing works inside a data center, not across them. Equally Spread kept processing flat (0.55 ms average) even where a data center carried double the traffic, but it could not send work to DC6, which sat idle and still cost $9.60. Routing between data centers is the service broker's job, and idle capacity is paid for either way.
- Always report tail latency (max / 95th percentile), not just the average: the average masked the worst failure mode in this study.